Variables, constants, operators, input, output and assignment

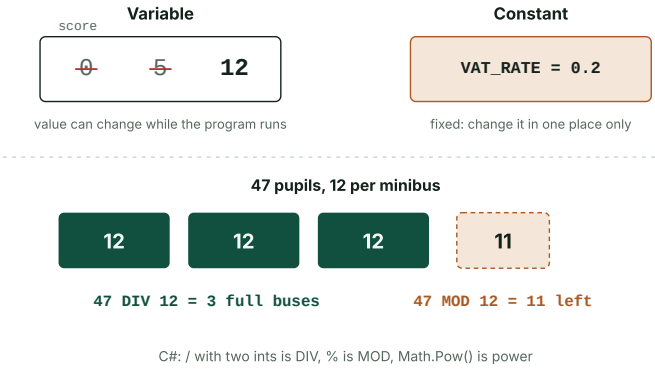
D14 FACT FILE

Specification: students should be able to

- Demonstrate understanding of and use the functionality of variables, constants, and Boolean and arithmetic operators
- Input, output and assignment statements

CCEA fact file: Programming Constructs 1

Key words: variable, constant, identifier, assignment, input, output, arithmetic operator, DIV, MOD, exponent, concatenation



A variable whose value changes compared with a constant, and DIV and MOD

Variables and constants

A variable is a named memory location whose value can change during execution; it should be initialised before use and given a meaningful identifier. A constant is a named value fixed for the whole run. Constants make code easier to read, mean a value only needs changing in one place, and stop the value being altered by accident.

Statements and operators

Input statements read data from the user, output statements display results and assignment stores a value in a variable. Arithmetic operators include +, −, *, /, DIV (whole-number division), MOD (remainder) and powers. Relational operators compare values and Boolean operators combine conditions.

T Key terms from the mark schemes

Variable
A named memory location whose value can change while the program runs.
Credited: U4 2019 Q2, 2025 Q3

Constant
A named value that stays the same while the program runs.
Credited: U4 2019 Q2, 2023 Q3, 2025 Q6

C C# example

Input, output, assignment

```
const double MILES_TO_KM = 1.609;          // constant
Console.Write("Name: ");
string name = Console.ReadLine();          // input
Console.Write("Miles walked: ");
double miles = Convert.ToDouble(Console.ReadLine());
double km = miles * MILES_TO_KM;           // assignment
Console.WriteLine(name + " walked " + Math.Round(km, 2) + " km");   // output
```

Exam focus

Identify variables and constants (2019 Q2(a)), compare them (2019 Q2(b)), define a constant and give a benefit (2023 Q3(b)), two benefits of constants (2025 Q6(c)). Algorithm questions expect correct assignment and output statements using the variable names given.

Unit 4 2025 Q6	[25]	Tent hire algorithm; Boolean; constants; validation true/false; Y/N validation loop
Unit 4 2023 Q3	[12]	Data types; constants and a benefit; sales total and average algorithm
Unit 4 2019 Q2	[13]	Variables and constants; miles-to-km algorithm with running total
Unit 4 2019 Q1	[18]	Data types, range and presence checks, extending a program, IDE features, translation

? Check yourself

1. Describe the difference between a variable and a constant. [2]
2. Give two advantages of using a constant in a program. [2]
3. What is the result of 23 MOD 5 and 23 DIV 5? [2]

Answers: 1. A variable is a named memory location whose value can change while the program runs (1) / a constant is a named value that does not change while the program runs (1) 2. Any two from: improves readability – a meaningful name instead of a 'magic number' (1) / easier maintenance – the value only needs changing in one place (1) / prevents the value being changed by mistake while the program runs (1). Do not credit a definition of a constant. 3. 23 MOD 5 = 3 (1) / 23 DIV 5 = 4 (1)